Retrieval-Augmented Generation

A Large Technical Knowledge Base for RAG Testing

Designed for chunking, embeddings, hybrid search, reranking, citations, and retrieval evaluation.

1. Introduction to RAG

Retrieval-Augmented Generation (RAG) is an architecture in which an application retrieves relevant information from a knowledge base before asking a language model to generate an answer. Instead of placing an entire document inside every prompt, the system finds a small set of useful passages and sends only those passages to the model.

Why RAG is useful

This section discusses why rag is useful in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

RAG pipeline

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Knowledge ingestion

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Query processing

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Answer generation

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Component	Purpose	Typical concern
Ingestion	Extract and normalize content	Bad extraction / OCR
Chunking	Split content into retrievable units	Too small or too large
Embedding	Represent semantic meaning	Model quality
Keyword search	Find exact terms	Tokenization
Vector search	Find semantic matches	Index and latency
Reranker	Refine candidate relevance	Compute cost
Generator	Produce grounded response	Hallucination

2. Document Ingestion

Document ingestion is the first major stage of a RAG system. Files such as PDFs, DOCX documents, HTML pages, Markdown files, and plain text are converted into machine-readable content. A production ingestion pipeline should preserve useful metadata such as document ID, filename, page number, section title, source URL, and chunk index.

PDF extraction

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OCR fallback

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Metadata

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Duplicate detection

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Incremental ingestion

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3. Chunking Strategies

Chunking divides long documents into smaller passages that can be indexed and retrieved independently. Chunk size affects retrieval precision, context completeness, embedding quality, storage cost, and downstream LLM performance. Sentence-based, paragraph-based, recursive, semantic, and parent-child strategies are commonly used.

Fixed-size chunks

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Sentence chunks

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Overlap

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Semantic chunking

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Parent-child retrieval

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4. Embeddings

An embedding model converts text into a numerical vector. Texts with similar meanings tend to have vectors that are close according to a similarity function. Embeddings are useful for semantic retrieval, but they can struggle with exact identifiers, short strings, unusual error codes, and highly structured values.

Vector representations

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Cosine similarity

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Embedding dimensions

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Model selection

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Embedding limitations

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5. Vector Databases

A vector database stores embeddings and provides efficient similarity search. PostgreSQL with pgvector is a practical option when an application already uses PostgreSQL. Common indexes include IVFFlat and HNSW. Vector search returns candidates that are semantically close to the query.

pgvector

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IVFFlat

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HNSW

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Distance metrics

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Metadata filtering

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6. Keyword Search

Keyword retrieval is especially strong when users ask about exact names, product codes, IDs, error messages, acronyms, or technical terms. Traditional search can complement semantic search because a vector model may treat visually similar strings as unrelated or may lose important exact-match information.

BM25

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Tokenization

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Term frequency

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Inverse document frequency

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Exact matching

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7. Hybrid Search

Hybrid search combines lexical and semantic retrieval. A typical system runs a keyword search and a vector search independently, then merges their rankings. Reciprocal Rank Fusion (RRF) is a simple and effective approach because it uses ranks instead of requiring the two systems to produce comparable raw scores.

Vector + BM25

This section discusses vector + bm25 in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

RRF

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Candidate pools

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Score fusion

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Failure cases

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8. Reciprocal Rank Fusion

RRF assigns a contribution to each result based on its rank. A common formula is score(d) = sum(1 / (k + rank(d))) across retrieval systems. A document appearing near the top of several independent result lists receives a high fused score.

RRF formula

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Choosing k

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Combining rankings

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Handling missing results

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Implementation notes

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9. Reranking

Reranking is a second-stage retrieval process. The first stage may retrieve 30 or 50 candidates cheaply. A more expensive cross-encoder or reranker then examines the query and each candidate together and produces a refined relevance score. The final five or ten chunks can be passed to the LLM.

Candidate retrieval

This section discusses candidate retrieval in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Cross-encoders

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Top-k selection

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Latency trade-offs

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Quality improvements

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10. Query Processing

User questions often need preprocessing before retrieval. Query processing can include normalization, language detection, query expansion, spelling correction, metadata filters, and decomposition of complex questions. The objective is to create a retrieval query that represents the information need accurately.

Query normalization

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Query expansion

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Filters

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Multi-query retrieval

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Question decomposition

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11. Context Construction

After retrieval, the application constructs the context supplied to the language model. Context should be relevant, compact, and traceable. Duplicate passages should be removed, and chunks can be ordered according to relevance or document structure.

Context limits

This section discusses context limits in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Deduplication

This section discusses deduplication in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Ordering

This section discusses ordering in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Metadata

This section discusses metadata in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed,

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Source attribution

This section discusses source attribution in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Component	Purpose	Typical concern
Ingestion	Extract and normalize content	Bad extraction / OCR
Chunking	Split content into retrievable units	Too small or too large
Embedding	Represent semantic meaning	Model quality
Keyword search	Find exact terms	Tokenization
Vector search	Find semantic matches	Index and latency
Reranker	Refine candidate relevance	Compute cost
Generator	Produce grounded response	Hallucination

12. Prompt Engineering for RAG

A RAG prompt should clearly define how the model uses retrieved evidence. Strong instructions can require the model to answer only from the provided context, state when evidence is insufficient, avoid inventing facts, and preserve citations. The prompt should separate system instructions from user content.

Grounding

This section discusses grounding in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Citation instructions

This section discusses citation instructions in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Abstention

This section discusses abstention in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Prompt injection

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inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Few-shot examples

This section discusses few-shot examples in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

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13. Hallucination Control

Hallucination occurs when a model produces unsupported or incorrect information. Retrieval quality is one important control, but retrieval alone is not sufficient. The generation layer should be instructed to distinguish evidence from assumptions and to refuse unsupported claims when the context does not contain an answer.

Grounded answers

This section discusses grounded answers in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

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Evidence checking

This section discusses evidence checking in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Confidence

This section discusses confidence in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model

inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Answer verification

This section discusses answer verification in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

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14. Citations

Citations allow users to trace an answer back to its evidence. Useful citation metadata includes document name, page number, section, chunk ID, and source location. Exact citations make debugging easier because developers can inspect the retrieved text that influenced an answer.

Page citations

This section discusses page citations in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Chunk IDs

This section discusses chunk ids in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Source metadata

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Traceability

This section discusses traceability in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed,

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Citation validation

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15. Security in RAG

RAG systems process untrusted documents and user queries. Retrieved content should not automatically be treated as instructions. Prompt injection can occur when a malicious document contains text designed to manipulate the language model. Applications should clearly distinguish data from executable instructions.

Prompt injection

This section discusses prompt injection in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Untrusted documents

This section discusses untrusted documents in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Access control

This section discusses access control in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Tenant isolation

This section discusses tenant isolation in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model

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Sensitive data

This section discusses sensitive data in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

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16. API Design

A RAG backend is commonly exposed through APIs. FastAPI is a Python framework suitable for defining typed request and response models. Typical endpoints include document upload, document listing, search, question answering, health checks, and authentication.

REST endpoints

This section discusses rest endpoints in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Request validation

This section discusses request validation in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Pydantic

This section discusses pydantic in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Error handling

This section discusses error handling in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed,

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API versioning

This section discusses api versioning in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

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17. Testing

Testing should cover ingestion, extraction, chunking, retrieval, ranking, API behavior, and answer generation. Unit tests can isolate individual functions, while integration tests can verify database and model interactions. pytest and mocking tools can make automated tests repeatable.

pytest

This section discusses pytest in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Mocks

This section discusses mocks in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Fixtures

This section discusses fixtures in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Integration tests

This section discusses integration tests in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed,

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Regression tests

This section discusses regression tests in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

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18. Caching

Model loading can be expensive because embedding and reranking models may occupy significant memory and require initialization time. Caching allows a process to reuse initialized model objects instead of loading them repeatedly. Python's lru_cache is one simple mechanism for process-level caching.

Model caching

This section discusses model caching in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

lru_cache

This section discusses lru_cache in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Cache lifetime

This section discusses cache lifetime in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Memory usage

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Invalidation

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19. Observability

Observability makes a RAG system measurable and debuggable. Useful telemetry includes request latency, retrieval latency, number of candidates, selected chunks, model latency, token usage, errors, and user feedback. Traces can connect one request across ingestion, retrieval, reranking, and generation.

Logging

This section discusses logging in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Metrics

This section discusses metrics in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Tracing

This section discusses tracing in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Latency

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excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Cost tracking

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20. Evaluation

RAG evaluation should measure both retrieval and generation. Retrieval metrics can include recall@k, precision@k, and mean reciprocal rank. Generation evaluation can consider faithfulness, answer relevance, completeness, citation correctness, and refusal behavior when evidence is missing.

Recall@k

This section discusses recall@k in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Precision@k

This section discusses precision@k in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

MRR

This section discusses mrr in the context of a production RAG application. A practical implementation should keep responsibilities separated so that ingestion, retrieval, ranking, generation, and API layers can be tested independently. The system should preserve identifiers and metadata throughout the pipeline. When a result is retrieved, developers should be able to determine which document and chunk produced it. Performance should also be measured because a technically correct approach can become impractical if database queries, embedding generation, reranking, or model inference introduce excessive latency. A robust implementation therefore balances accuracy, speed, memory usage, maintainability, and security.

Faithfulness

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Answer relevance

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Appendix A — Retrieval Test Data

Field	Value
Document ID	DOC-2026-RAG-001
Chunk ID	CHK-00482
Error code	ERR_VECTOR_TIMEOUT_504
API endpoint	POST /api/v1/query
Model	all-MiniLM-L6-v2
Reranker	cross-encoder/ms-marco-MiniLM-L-6-v2
Database	PostgreSQL + pgvector
Search	BM25 + vector + RRF
Index	IVFFlat
Configuration	CHUNK_OVERLAP_PERCENT=10

Appendix B — Sample Questions

• What is the difference between BM25 and vector search?

• Why does hybrid search improve retrieval?

• How does Reciprocal Rank Fusion combine rankings?

• Why is chunk overlap useful?

• What is the purpose of a reranker?

• How can citations reduce debugging difficulty?

• What happens when the answer is not present in the retrieved context?

• Why should untrusted document text not be treated as instructions?

• What metrics can be used to evaluate retrieval quality?

• Why can lru_cache help when loading embedding models?